技术要求



1.封面封底 157g 铜版纸覆哑膜彩打，内部纸 80g 双胶纸黑白印刷，正反打印；
2. 装订方式：80 页，胶装；
3. 未注尺寸公差按 ±1.5mm；
4. 图面、字体印刷清晰、无偏移 / 无毛边、不起边、油墨不脱落；
5. 字体颜色为 PANTONE Black C，无边框，底色为白色；
6. 符合 RoHS 要求。

物料描述	User Manual_Motion Compact DC WallBox_EN_Livoltek		设计	卢柯安	2025/03/20
			审核	刘康楠	2025/03/20
料号			核准		
材质	NA	单位	mm	杭州利沃得电源有限公司	

LIVOLTEK

USER MANUAL



Copyright Declaration

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About This Manual

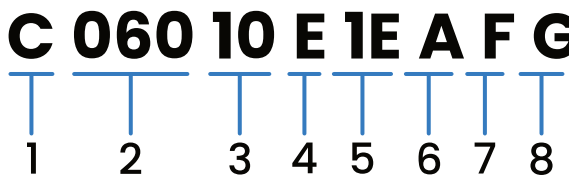
Validity

This manual is an integral part of Motion Compact DC WallBox. It provides instructions for installation, electrical connection, commissioning, maintenance and troubleshooting of the product. Please read it carefully before operating.

This manual is valid for the following models:

- C06010E1EAFG
- C06010E2EAFG
- C06010EEUAFG
- C06010E1EAFF
- C06010E2EAFF
- C06010EEUAFF
- C04010E1EAFE
- C04010E2EAFE
- C04010EEUAFE
- C03010E1EAFD
- C02010E1EAFB
- C01510E1EAFA

Model Name Description



No.	Meaning	Description
1	Product Name	"C": Motion Compact DC WallBox
2	Power	Nominal output power, Value range: 015 (15 kW), 020 (20 kW), 030 (30 kW), 040 (40 kW) and 060 (60 kW).
3	Output Voltage	"10": 1000 V DC output Voltage.
4	Standard	"E": IEC Standard.
5	Connector set	1E(CCS2), 2E(CCS2 + CCS2), EU (CCS2 + CCS1).
6	Generation	"A": First-generation product.
7	Power module	"F": ACE European standard 30 kW power module.
8	Maximum output current	"A": 50 A, "B": 65 A, "D": 100 A, "E": 133 A, "F": 150 A, "G": 200 A.

Target Group

Quality requirements for installation and maintenance personnel:

Have the qualification certificate or experience in electrical power system installation and maintenance, and have the qualification to engage in relevant work (such as live working). In addition, they must have the following knowledge and operating skills.

- Have basic knowledge and installation skills of EV charger.
- Have basic knowledge and operation skills of electrical connection.
- Have basic power safety knowledge and skills, and are able to read the contents of this manual.

Symbol Description

When installing, operating and maintaining the equipment, please read this manual first and follow all the safety precautions marked on the device and in the manual. For a better using experience and to protect personal and property safety, please read the following symbols carefully.



DANGER

Indicates a hazardous situation which, if not avoided, will or could result in death or serious injury.



WARNING

Indicates a moderate or potential hazard situation which, if not avoided, will or could result in death or serious injury.



CAUTION

Indicates a low level of hazard situation which, if not avoided, will or could result in death or serious injury.



NOTE

Provides additional information to emphasize or supplement important points of the main text.

Manual Preservation

Keep this manual well for future reference.

Release Note

Version	Date	Update Record
V1.0	2025-03-20	Initial release.
V1.1	2025-07-22	Replaced some pictures, polished some expressions and updated Troubleshotting.

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1 Safety

1.1 Statements

DANGER

Persons not authorized by LIVOLTEK are not allowed to open the charger.
Do not disassemble, repair or refit the charger and relevant ancillary facilities without permission. Improper operation may cause damage, water leakage, electric leakage and other damage to the charger or potential personal injury. If you need maintenance, modification, etc., please contact LIVOLTEK customer service.
Do not put flammable and explosive materials, chemicals, and other dangerous materials near the charger.
Children are not allowed to approach or use the charger during charging to avoid accidental injury.

WARNING

Please start and stop charging according to the user manual and relevant prompts provided by LIVOLTEK. When plugging in and out the charging connector, pay attention to applying appropriate force and press the unlock button.
When inserting the charging connector, ensure that the lock catch of its head is fully fastened to the groove of the vehicle inlet to ensure full connection; do not pull the connector too hard, or pull it forcibly when it is in locked state.
It is forbidden to move the vehicle during charging, and it is forbidden to draw the charging connector during charging.

WARNING

In case of emergencies such as leakage, fire, electric shock, press the emergency stop button immediately.
After the charging operation is completed normally, please return the charging connector back into the socket in time to prevent contamination.
Use this product only after confirming it is safe.
In wet weather, it shall be confirmed that the charging connector and vehicle inlet are dry. If the charger or charging connector is found to be wet, charging shall be prohibited.

DANGER

It is strictly prohibited to use the charger if it (including the charging connector, cable, leakage protection device, and other auxiliary parts) shows defects, cracks exposed wires, etc. Contact LIVOLTEK customer service immediately in case of any concerns.

It is strictly forbidden to touch the charging connector and vehicle inlet with metal objects. If the charging connector and vehicle inlet are found to be damaged by foreign matters, please stop using the charger immediately and consult the LIVOLTEK customer service for proper treatment.

WARNING

For the unused charger, it is recommended to disconnect the circuit breaker associated with the charger to prevent power loss or unknown accidents.

Adaptors or conversion adapters are not allowed to be used.

1.2 Symbols on the Label

Table 1-1 Symbols

Symbol	Description
CE	CE Mark.
RoHS	RoHS Mark.
	Potential risks exist. Wear proper Personnel Protective Equipment before any operations.
	HIGH VOLTAGE HAZARD
	WEEE Mark. Do not dispose of the charger together with household waste.

2 Products Introduction

The Motion Compact DC WallBox is a high power charging equipment for fast EV charging needs. With stable output power and high charging efficiency, this charger provides fast, safe and reliable charging services for electric vehicles.

2.1 Appearance



Figure 2-1 MC Series (Pole-mounted)



Figure 2-2 MC Series (Wall-mounted)

2.2 Performance and Features

- The system has a modular design with parallel outputs for flexible configuration and easy maintenance.
- The system provides comprehensive protection.
- The appearance is simple and beautiful.
- Adopts high-definition touch screen with vivid colours and IP55 level of protection.
- Supports multiple communication modes.
- The system supports OCPP1.6 and will be upgraded to OCPP2.0.1 in the future.

2.3 Appearance

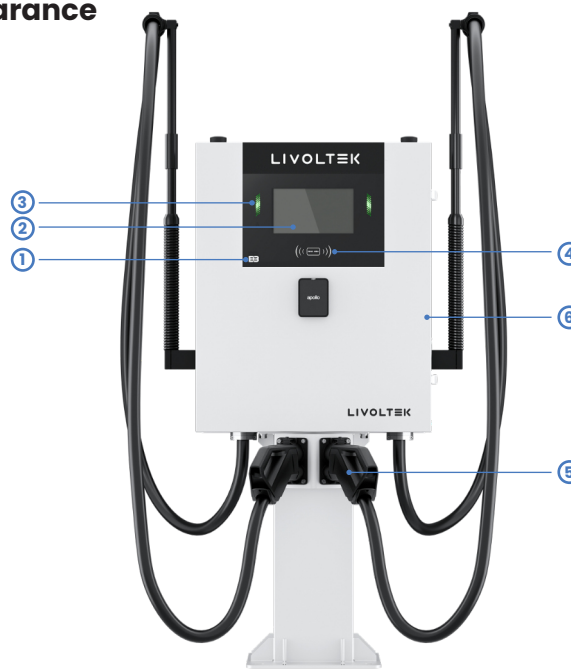


Figure 2-3 MC Series

Table 2-1 Charger Operational Description

No.	Name	Description
①	Power indicator	White light remains steady on when the device is powered on.
②	Display touch screen	HMI interaction
③	Status indicator	Device running status
④	Swipe card area	Gently place the RFID card on this area and swipe to start or stop charging.
⑤	Charging connector holder	Insert the charging connector to the charging connector holder after use to prevent damage.
⑥	Emergency stop button	Press to stop charging output immediately in case of system abnormalities.

3 Transportation and Storage

Transportation

- During transportation, pay attention to the warning signs on the packing case and do not subject it to violent impact.
- The device should be placed strictly according to the direction marked on the packing case to avoid vibration damage to the device.
- Do not ship this device with flammable, explosive and corrosive articles.
- Do not expose this device to snow, rain or other liquid substances and mechanical damage.

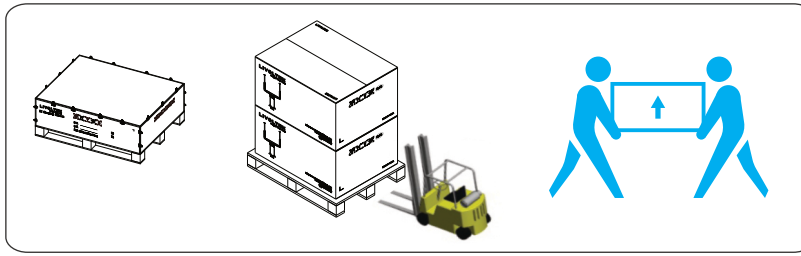


Figure 3-1 Transportation

Storage

- Store the device in accordance with the directional indicators marked on the packing case.
- Ensure the packing case is positioned at least 200 mm above the ground and maintains a minimum clearance of 500 mm from walls, heat and cold sources, windows, or air inlets.
- Keep the device from any kinds of hazardous gases, flammable, explosive substances and
- corrosive chemicals.
- Keep the device from strong mechanical vibration, impact and magnetic field effect.
- The relative humidity of the storage environment should be between 20% and 80%.
- Inspect the stored device every six months.

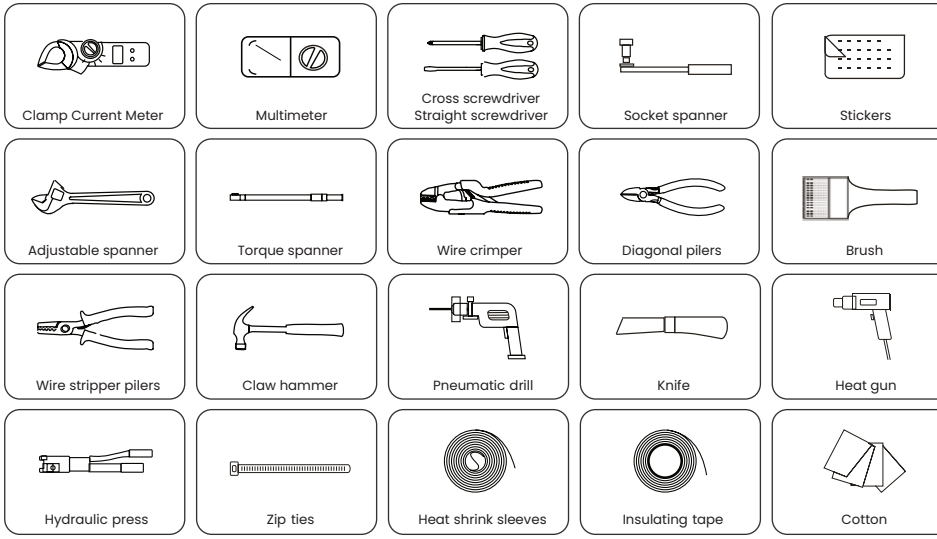
4 Installation Preparations

4.1 Installation Requirements

4.1.1 Environment Requirements

- This device is an outdoor type electric vehicle charger, which meets an ingress protection rating of IP55 and is suitable for installation in dry and dust-free environments.
- The instllation foundation must be stable enough to bear the device weight
- Ensure the installation environment temperature is within -30 °C to +55 °C to improve device performance.
- Ensure the installation environment is well ventilated and away from water sources, heat sources, flammable and explosive matters.
- Keep the device from direct sunlight, dust, volatile gas, any corrosive substances and excessive salt.
- When installed in an open environment, consider install a canopy above the device to protect it from direct rainwater drenching.

4.1.2 Tools Requirements



4.1.3 Required Materials

Refer to Table 4-1 to see the cable diameter specifications.
When the cable length exceeds 200 m, considering the pressure drop on the cable, and increase the cable diameter accordingly.

Table 4-1 Recommended Cable Specification

Power	15 & 20 & 30 kW		40 & 60 kW	
	A/B/C/N	Ground PE	A/B/C/N	Ground PE
Wire Size [AWG]	5	5	2	5
Terminal Model	SC16-8	SC16-8	SC35-8	SC16-8

4.1.4 Pole-Mounting Cable Length Requirements

When pole-mounting, ensure both power and communication cables have sufficient length for future wiring, with at least 1 m of power cable and 2 m of communication cable extending beyond the installation base.

5 Unpacking and Inspection

5.1 Unpacking

- This device undergoes 100% testing and inspection before delivery. However, damages may still occur during transportation. Before unpacking, please carefully check the external packaging for any signs of damage like punctures or cracks.
- Move the device with the package to the installation site with a folklift. Two people are recommended to carry the device and put the device on the ground. For pole-mounted chargers, the pole is packed separately.
- Unpacking according to the following figures.

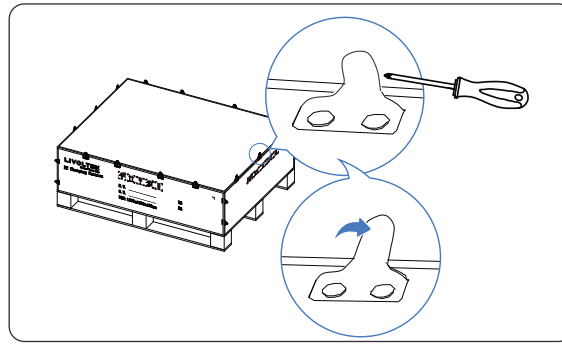


Figure 5-1 Unpacking Charger Package

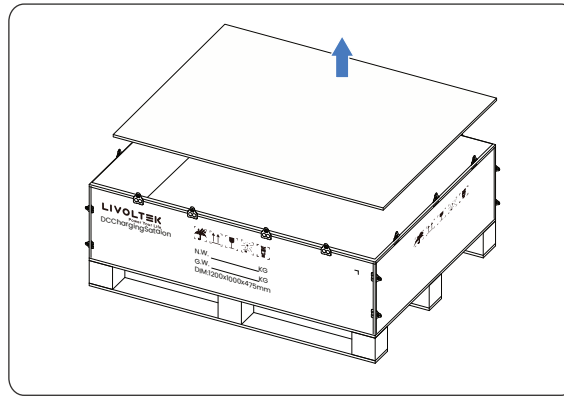


Figure 5-2 Taking Out Charger Accessories

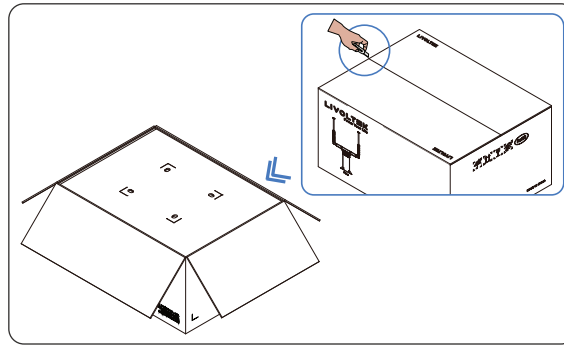


Figure 5-3 Unpacking Pole Package

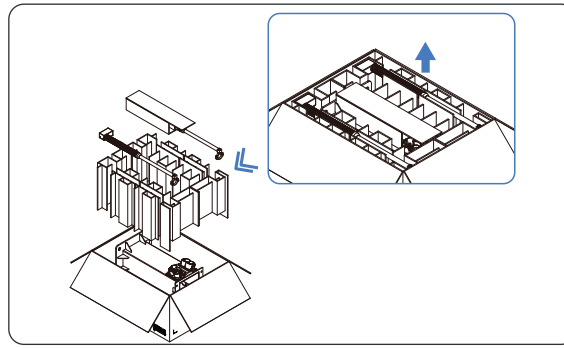


Figure 5-4 Taking Out Pole Accessories

Properly handle all the packaging materials for future storage or transportation.

Upon opening the package, check whether the Charger is intact and whether all accessories are included. If any damage is found or any accessories are missing, contact your dealer immediately.

CAUTION

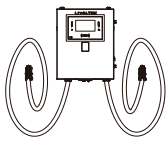
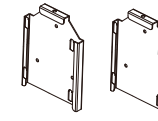
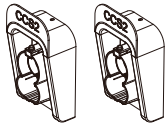
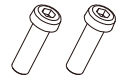
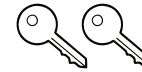
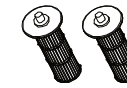
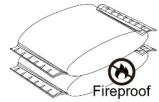
- keep the forklift steady when lifting the device.
- wear protective gloves when unpacking.

5.2 Packing List

NOTE

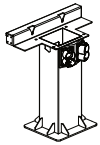
Packing list below is based on the double-connector charger. For the single-connector charger, the quantity of accessories marked with * is half that of the double-connector charger, while other accessories is the same.

Charger

A  Charger*1	B  Charger Mounting Bracket*1	C  Wall Hanging Accessory*4	D*  M6 × 16 Screw*2
E  M10 × 80 Expansion Bolt*4	F*  Ø 6 × 28 Plastic Anchor*6	G*  ST3.9 × 30 Self-tapping Screw*6	H*  Charging Connector Holder Mounting Bracket*2
I*  Charging Connector Holder*2	J  M5 × 12 Screw*2	K  Key*2	L  Handle*2
M  Trox Key*1	N  Fireproof Putty*2	O  Certification of Conformity*1	

Pole

A



Pole*1

B* Spring Cable Holder Set (optional)



Spring*2



Spring Base*2



M4*20 Screw*8



Cable Clamp*4



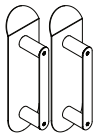
Clamp Liner*4

C



M6 × 100 Expansion Bolts*4

D* Handle-shaped Cable Holder Set (optional)



Handle-shaped Cable holder*2



M6 × 16 screws*4

E



M12× 25 Screw*6

F



Drilling Template*1

G



Certification of Conformity*1

6 Installation and Wiring

6.1 Wall-mounting Procedure

Mounting steps for single-connector charger is same as that of the double-connector charger, except that the single-connector charger only includes one cable holder. The mounting steps below is based on double-connector charger.

Before you start, ensure the device and all accessories are complete and intact.

Step 1: Install charger mounting bracket.

a): Put the charger mounting bracket on the wall, 600 mm above the floor, ensuring it is perpendicular to the ground.

b): Mark the mounting bracket's drilling position on the wall.

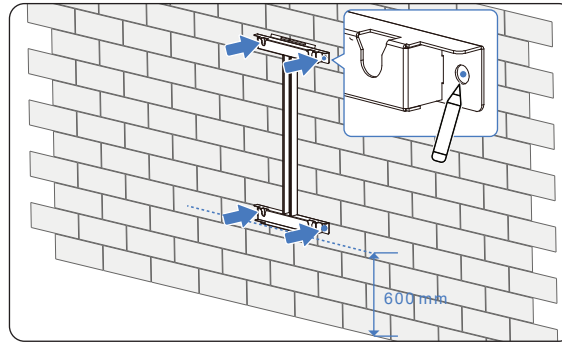


Figure 6-1 Marking Drilling Position

d): Drill 80 mm deep holes at the marked positions.

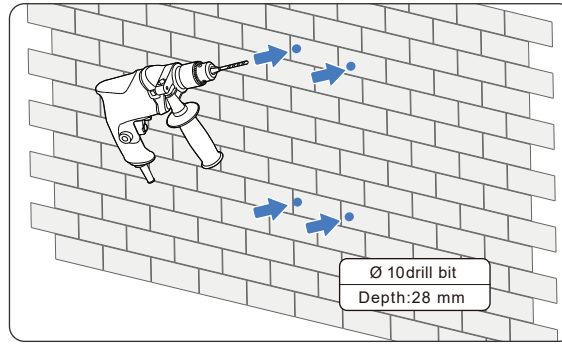


Figure 6-2 Drilling Holes for Charger Bracket

e): Secure the mounting bracket onto the wall with M10*80 expansion bolts.

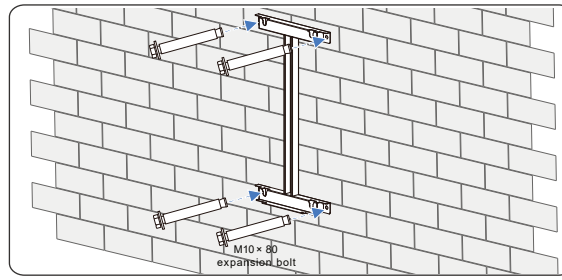


Figure 6-3 Securing the Bracket

Step 2: Install mounting brackets of the charging connector holder.

a): Put the connector holder mounting brackets symmetrically on both sides of the charger mounting bracket, 600 mm perpendicular to ground and 500 mm horizontal to the charger mounting bracket.

b): Mark the mounting bracket's drilling positions on the wall.

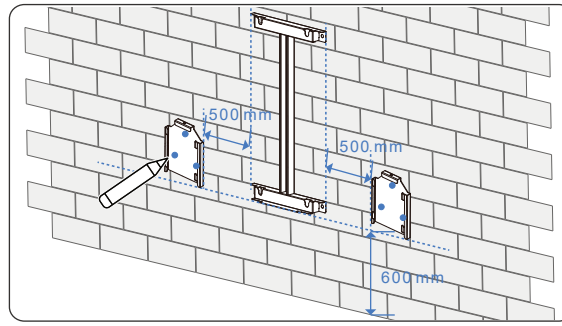


Figure 6-4 Marking Drilling Position

c): Drill 28mm deep holes at marked locations.

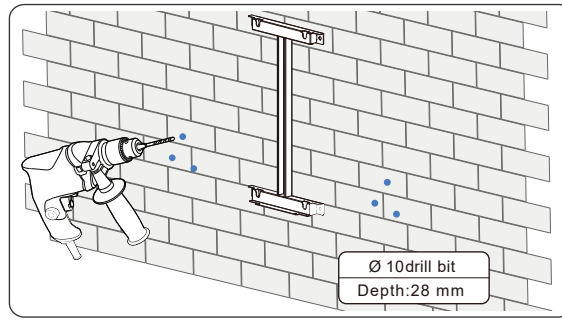


Figure 6-5 Drilling Holes for Charging Connector Holder Bracket

d):Drive the Φ6*28 plastic anchors into the drilled holes.

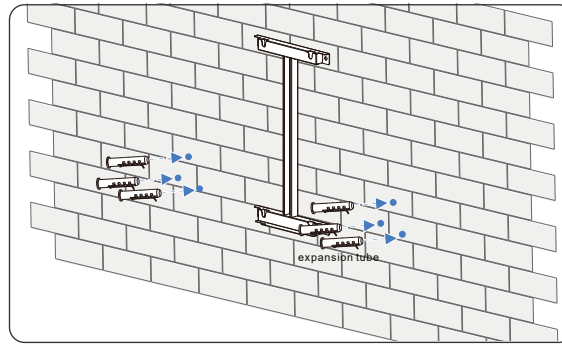


Figure 6-6 Inserting Expansion Tubes

e): Align mounting holes of the mounting brackets with the plastic anchors on the wall, and secure the mounting brackets on the wall with the ST3.9*30 self-tapping screws.

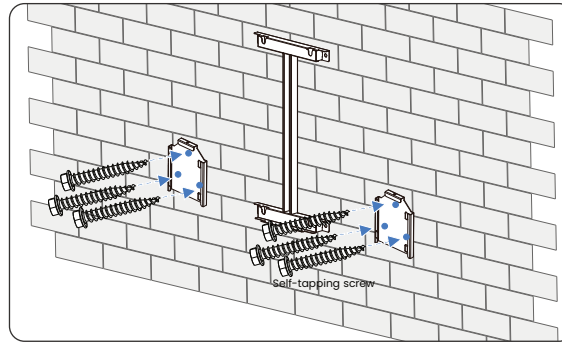


Figure 6-7 Inserting Self-tapping Screws

Step 3: Mount the charging connector holders to the installed mounting brackets with M5 × 12 screws.

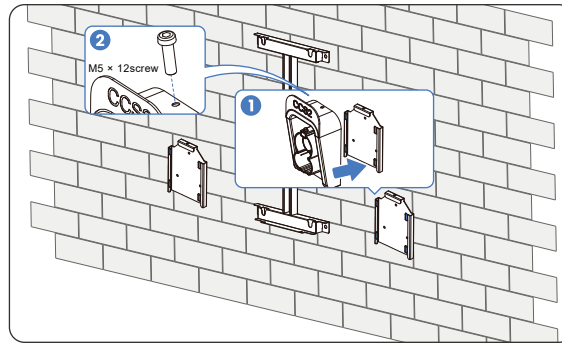


Figure 6-8 Assembling Charging Connector Holders

Step 4: Assemble the charger to the charger mounting bracket.

a): Remove the waterproof plugs from the charger (one on left and right, two at the bottom, and four at the back).

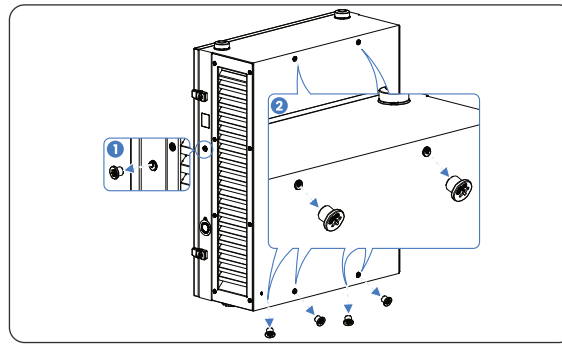


Figure 6-9 Removing Waterproof Plugs

b): Install the handles on both sides of the charger, and four wall hanging accessories at its back.

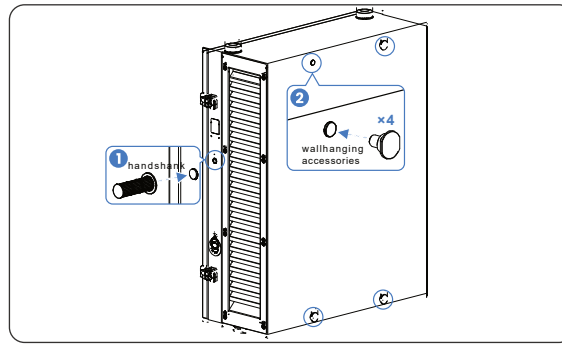


Figure 6-10 Installing Handles and Wall Hanging Accessories

c): Hang the charger onto the bracket, ensuring a secure fit.

NOTE

When moving the charger, a minimum of 2 people are needed.

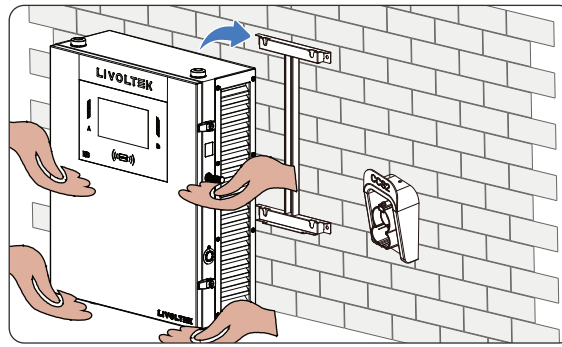


Figure 6-11 Mounting the Device

d): Secure the charger to the bracket with two M6 × 16 screws at the bottom of the charger.

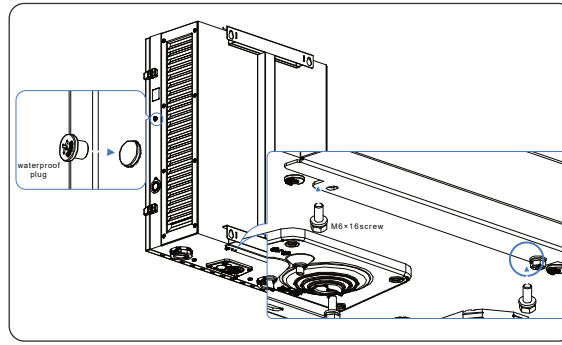


Figure 6-12 Securing the Device

d): Remove the handles and insert waterproof plugs back into the vacant holes.

Step 5: Put the charging connectors into the connector holders.

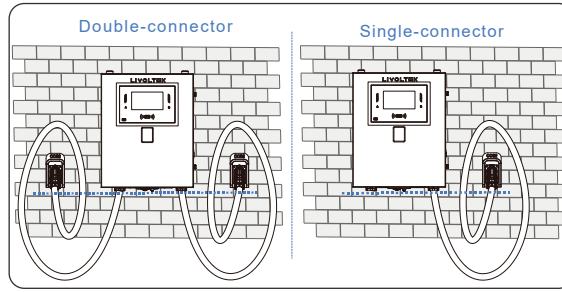


Figure 6-13 Same Installation Procedure

6.2 Pole-mounting Procedure

Pole-mounting steps for single-connector charger is same as that of the double-connector charger, except that the single-connector charger only includes one spring or handle-shaped cable holder. The mounting steps below is based on double-connector charger.

Before you start, ensure the device, the pole and all accessories are complete and intact.

Step 1: Prepare the installation base.

a): Make a cement base with dimensions of at least 470*430*100 mm (L*W*H) at the installation site.

b): Put the drilling template on the base and mark the drilling positions.

c): Remove the template and drill holes to a depth of 60 mm (Φ 16 mm).

d): Reserve a Φ 100 mm hole at the base center for the power cable.

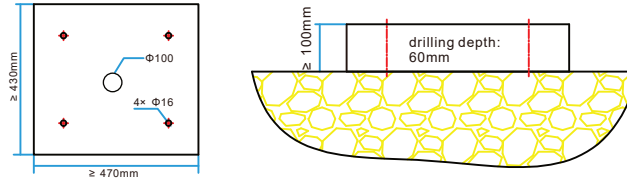


Figure 6-14 Preparing Installation Base

Step 2: Secure the pole to the base.

a): Align the four mounting holes at the pole bottom with the four Φ16 holes on the installation base.

b): Insert the M16×100 expansion bolts through the aligned holes, then secure them with a torque wrench.

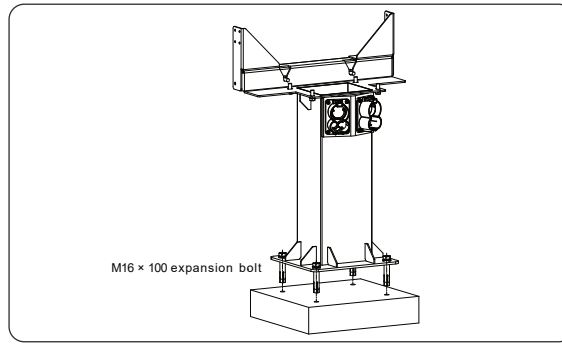


Figure 6-15 Securing the Pole

c): Open the back cover of the pole and route the power cable through the top of the pole.

d): Adjust the cable length, and re-install the back cover

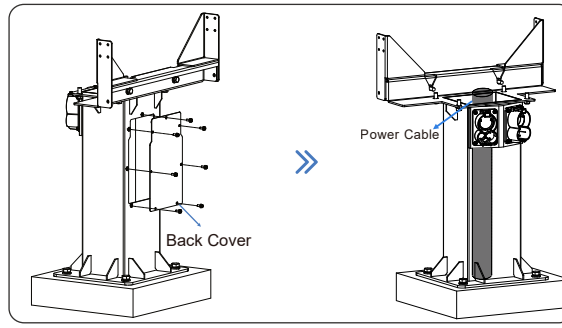


Figure 6-16 Leading Power Cable into Pole

Step 4: Assemble the charger to the pole

a): Cut open the cable inlet at the charger bottom.

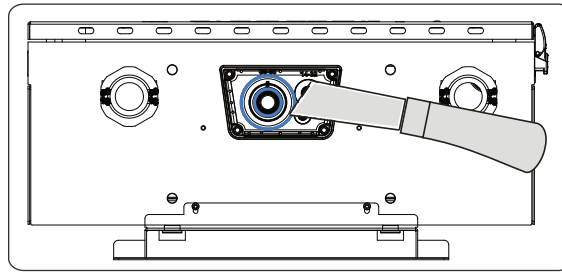


Figure 6-17 Cutting Cable Inlet

b): Remove the waterproof plugs from the charger: one on each side (left and right), four at the back, and two at the bottom).

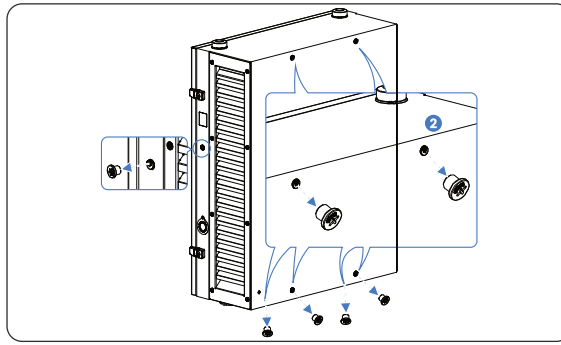


Figure 6-18 Removing Waterproof Plug

c): Attach the handles to both sides of the charger, then position the charger onto the pole until fully seated.

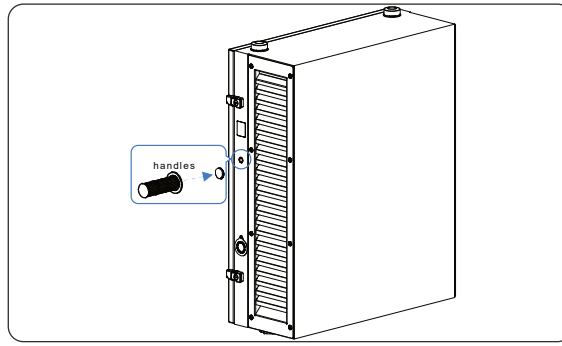


Figure 6-19 Installing Handles

d): Align the charger's mounting holes (two at the lower back and four at the bottom) with corresponding holes on the pole and secure the charger to the pole using M12*25 screws.

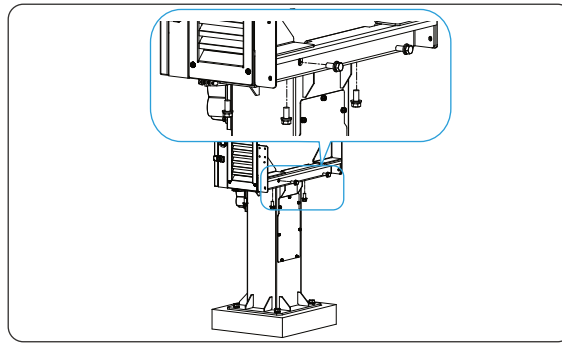


Figure 6-20 Securing Charger to Pole

e): Remove the handles and insert the waterproof plugs back into the vacant holes.

Step 4: Assemble the spring cable holder to the pole (optional, skip this step if using handle-shaped cable holder)

a): Assemble the springs to the spring bases.

b): Remove the M6*16 screws on the other side of the spring bases, then assemble the springs to the pole with the M6*16 screws.

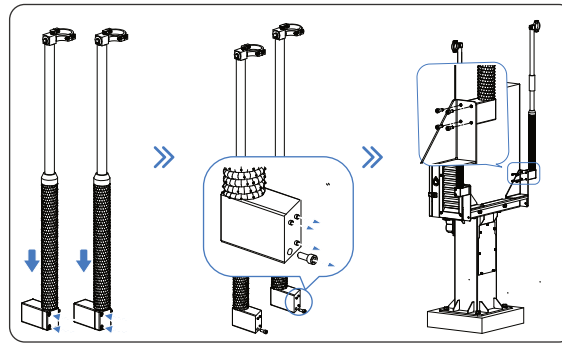


Figure 6-21 Securing the Springs

c): Open the metal bracket on top of the springs by removing the M6*16 screws on it.

d): Install the clamp liner and cable clamp around the charging cable with M4*20 screws, maintaining a 2.85 m distance between the charging connector and the clamp.

e): Place the cable's clamped section into the metal bracket on the spring and lock the bracket with M6*16 screws.

f): Put the charging connectors into the connector holders.

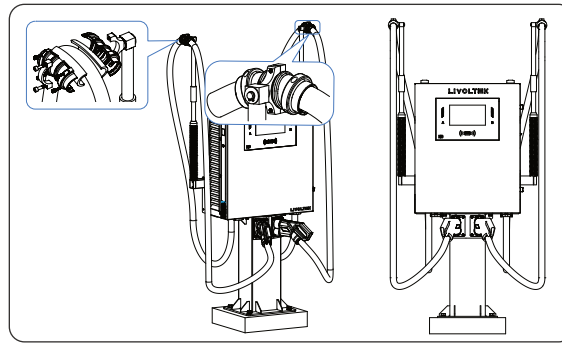


Figure 6-22 Assembling Charging Cables

Step 5: Assemble the handle-shaped cable holder to the pole (optional, skip this step if using spring cable holder).

a): Align the cable holders with the mounting holes on the pole, then secure them by installing the M6*16 screws.

b): Organize the cables neatly on the cable holder and put the charging connectors into the connector holders.

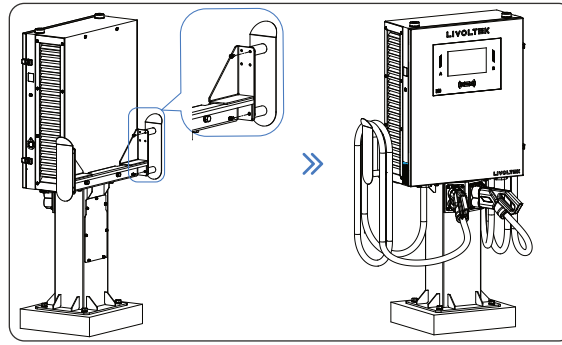


Figure 6-23 Assembling cable holder

6.3 Electrical Connection

NOTE

Ensure the device is free from electricity before you start.

Step 1: Open the door with the key.

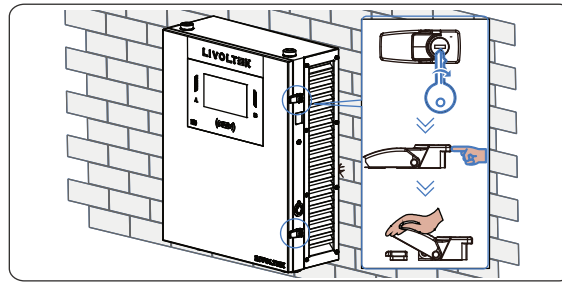


Figure 6-24 Opening the Door

Step 2: Route the power cable into the charger through the cable inlet.

Step 3: Strip 180 mm of the outer cable jacket to reveal the L1/L2/L3/N/PE wires.

Step 4: Cut 80 mm off the ends of the L1, L2, and L3 wires.

Step 5: Strip insulation of the L1/L2/L3/N/PE wires properly and crimp insulated terminals onto them.

Step 6: Remove the screws from the power terminals.

Step 7: Conenct the L1/L2/L3/N/PE wires to their designated terminals and secure with terminal screws (torque: 11 N·m).

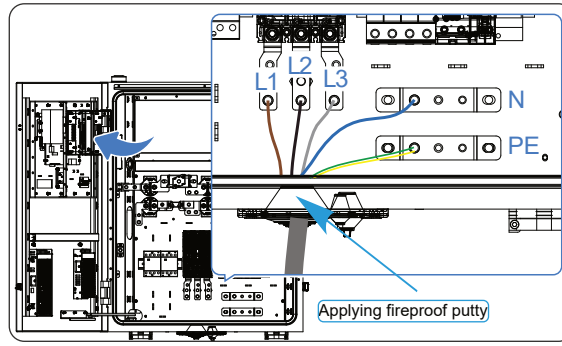


Figure 6-25 Wring

Step 8: Apply fireproof putty at the internal side of the cable inlet.

Step 9: Close and lock the door after wring is completed.

7 Operating Instructions

This chapter introduces the touch screen operation of the charger, including the touch screen function, homepage introduction, switching operation, screen setting, parameter setting, current fault query, history information query and restoring factory settings.

NOTE

Parameter values and other details seen in this chapter are for illustration only; please refer to the actual parameters of the received product.

7.1 Touch Screen Functions

The following functions can be realized by operating the LCD touch screen.

Table 7-1 Touch Screen Functions

Item	Function Description
Home Page	Users can check the current status of the charger: idle, charging in progress, charging completed.
Account login	Different personnel have different permissions for setting and modifying.
Operation Information	Display all the current measurement data and operation status of the system.
Charging Records	Users can view the charging records, including charging start and end time, payment card number, charging duration, charging amount, charging power, charging maximum current, voltage and vehicle identification number.
Network Settings	The network setting interface allows you to configure the server IP address, server port, local IP address, subnet mask, default gateway and terminal number.
Screen Settings	Users can set the date and time.

7.2 Home Page

When powered on, the system enters the homepage by default.

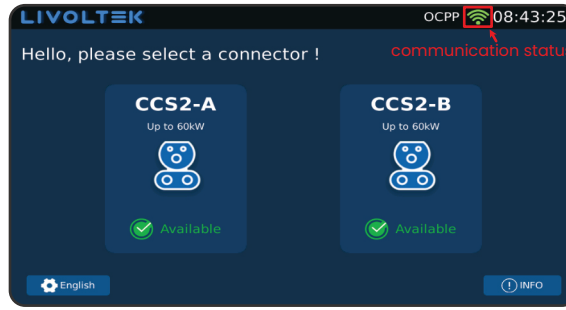


Figure 7-1 Homepage

The homepage provides information such as charging connector type and status, and communication status, etc.

NOTE

The Wi-Fi icon can have different colors, indicating different communication status. Check the table below for details.

Color	Communication Status
Green	connected to OCPP platform and communication normal
Blue	connected to OCPP platform but no data exchange detected
Red	disconnected to OCPP platform

7.3 Initial Setup

Step 1: Click **LIVOLTEK** located at the top left corner of the homepage to access the login page.

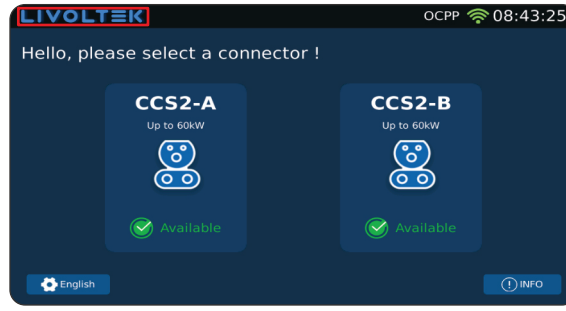


Figure 7-2 Home Page

Step 2: Select a role for **user** and enter the password to login.

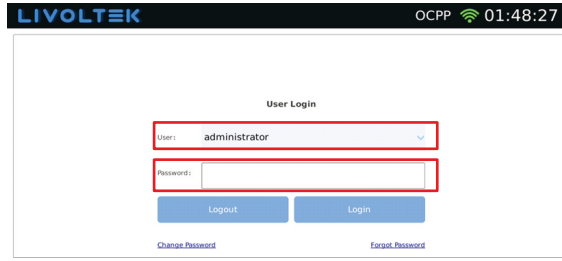


Figure 7-3 Login Page

NOTE

- User roles include "**operator**" and "**administrator**", the latter has more permissions.
- The default password for operator is "**111111**", and "**123456**" for administrator.
- Click **Change password** to set your own password.

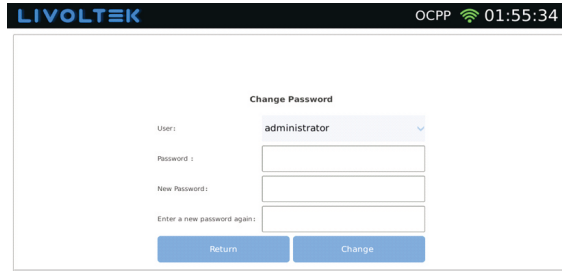


Figure 7-4 Changing Password

Step 3: Configure timezone.

a): After logging in, click **Syestem** → **Basic** on the left of the page.

b): Set the timezone.

c): Click **Save Change**.

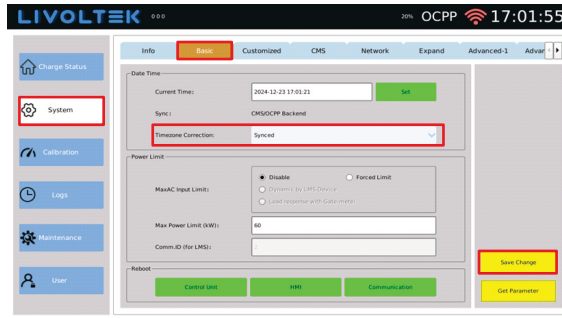


Figure 7-5 Configuring Time Zone

Step 4: Configure the network.

a): After logging in, click **Syestem** → **Network** on the left of the page.

b): Set the network mode.
c): Click **Save Change**.

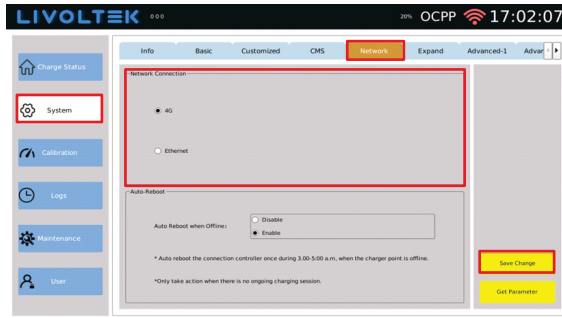


Figure 7-6 Configuring Network

NOTE

Refer to the picture below to locate the ethernet interface and the 4G SIM card slot.

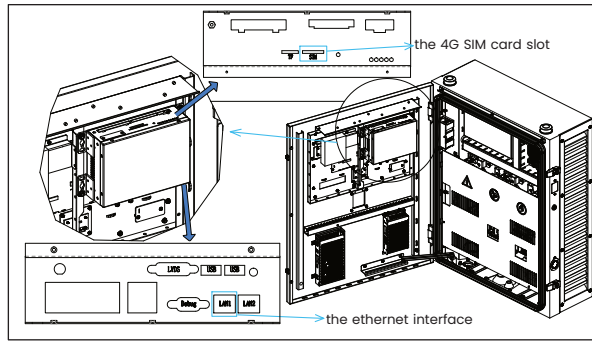


Figure 7-7 Ethernet Interface & 4G SIM Card Slot

Step 5: Configure the CMS.
a): After logging in, click **Syestem → CMS → View And Configure**.

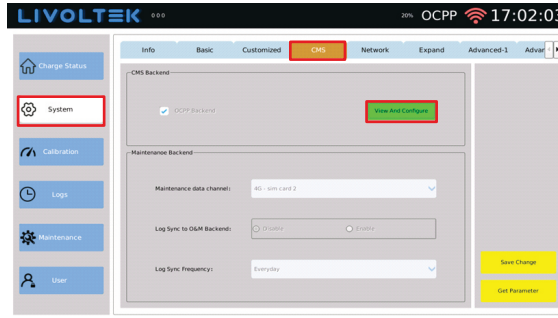


Figure 7-8 Clicking View And Configure

b): Input the **CMS URL**, which refers to your OCPP platform url.

c): Input the **Charge Point Identify**, which refers to your Charger ID on OCPP platform.

d): Select "0" or "1" for **Security Profile** when the url starts with "**ws**", select "2" when it starts with "**wss**" .

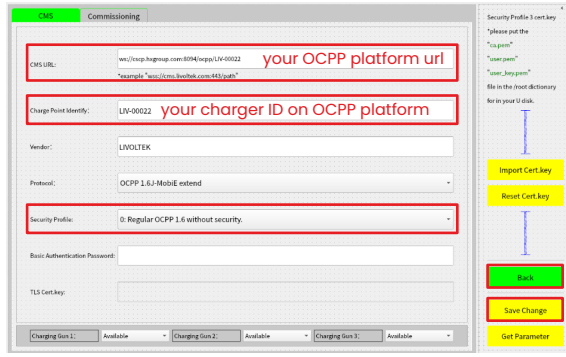


Figure 7-9 Configuring Security Profile

e): Click **Save Change→Back**.

f): Click **Commissioning → Connect Server→Back**.

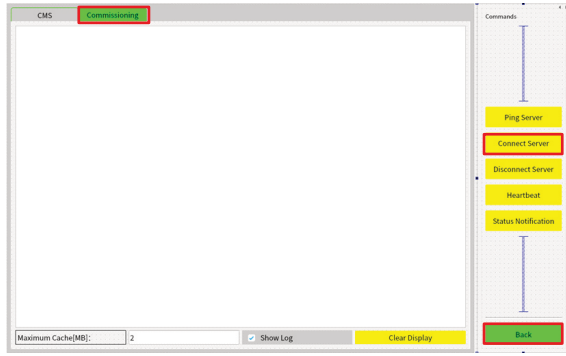


Figure 7-10 Connecting Server

Step 6: Configure the **Service Call** and **Lockscreen Image**.

a): After logging in, click **Syestem → Customized**.

b): Configure the **Service Call** and **Lockscreen Image**

c): Click **Save Change**.

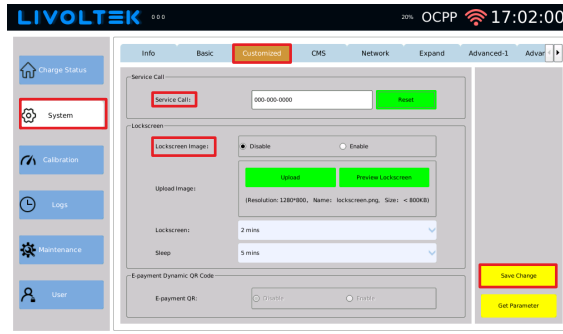


Figure 7-11 Configuring Customized

NOTE

When configuring Service Call, clicking **Reset** would reset the call to the default number of "000-000-0000" .

7.3 Charging Process

Step 1: Select the type of charging connector on the homepage.

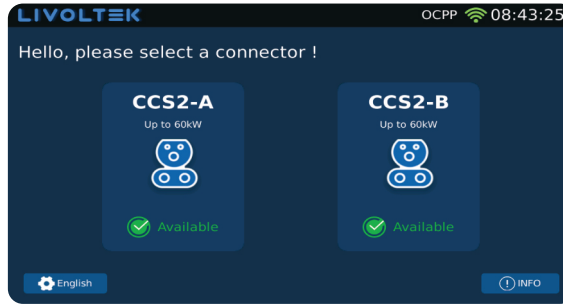


Figure 7-12 Home Page

Step 2: Select an authentication method and authenticate.

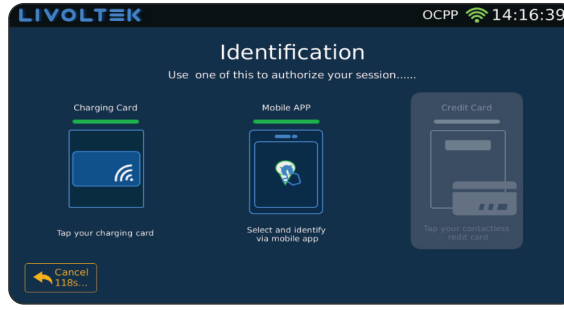


Figure 7-13 Authentication Method

Step 3: Insert the charging connector into the vehicle inlet to start charging.



Figure 7-14 Connecting Vehicle to Charging Connector

NOTE

Once charging begins successfully, the system automatically navigates to the charging details page, where users can monitor real-time charging information.



Figure 7-15 Charging Detail Page

Once charging has been in progress for some time, the screenautomatically reverts to the homepage.

Step 5: Click **End Session** on the home page and authenticate to end charging.



Figure 7-16 Home Page

NOTE

Once charging is complete, the system automatically shows a summary of the charging session for users to check.



Figure 7-17 Charging Session Sumary

Step 6: After charging ends successfully, disconnect the charging connector from the vehicle and return it to its holder.

7.4 LED Indicators

The status indicators consist of green, red and blue lights, indicating "standby", "charging" and "alarm" respectively.

Table 7-2 Status Indicator Description

Indicator Color	Description
Green	**Steady on:** powerd on and standby
Blue	**Flashing:** connector connected to vehicle but charging has not begun. **Steady on:** charging in process.
Red	**Flashing:** fault occurs while not charging. **Steady on:** fault occurs during charging.

8 Troubleshooting and Maintenance

8.1 Troubleshooting

Refer to the table below for specific faults and recommended actions.

Table 8-1 Troubleshooting list

Fault	Recommended Action
TCU and CCU communication failure	1. Verify that the communication wiring is secure. 2. Power cycle the device.
TCU and platform offline warnings	1. Verify that the network connection is stable. 2. Power cycle the device. 3. Try other network connection methods.
Communication failure between TCU and energy meter	1. Verify that the communication wiring is secure. 2. Replace the TCU/Meter. 3. Power cycle the device.
TCU and card reader communication warning	1. Verify that the communication wiring is secure. 2. Replace the TCU/Card Reader.
Emergency stop button action fault	1. Verify that the emergency stop button is not in the pressed state. 2. Verify that the wiring is secure. 3. Power cycle the device. 4. Replace the emergency stop button.
AC input circuit breaker fault	/
DC bus bar output contactor refuses to act/mis-actuates faults	1. While not charging, short the control terminal of the output DC contactor to check whether it operates normally. If not, replace the contactor. 2. Power cycle the device and retry charging process.
Lightning arrester fault	1. Check whether the surge protective device has turned red. If not, check if the wiring is secure or whether the voltage at the remote signaling contact and the continuity of the contactor are normal; if red, replace the surge protective device.
Insulation monitoring fault	1. Power cycle the device and retry charging process.
Vehicle battery reverse fault	/

Fault	Recommended Action
Vehicle control guide failure during charging	1.Power cycle the device and retry charging process.
Charging pile overheating fault	1. Check whether the resistance of the air inlet and outlet temperature sensors matches the PT1000 resistance table. If yes, check whether the wiring is secure; if not, replace the temperature sensor (disconnect the wiring before measuring resistance). 2. Power cycle the device and run charging process for approximately 30 minutes to see if the fault is solved.
BMS communication fault	1. Power cycle the device and retry charging process.
Input voltage overvoltage fault	1.Verify that the input voltage is below the upper limit (537V). 2. Power cycle the device and re-insert the charging connector.
Input voltage undervoltage fault	1. Verify that the input voltage is above the lower limit (323V). 2. Power cycle the device and re-insert the charging connector.
DC bus output overvoltage fault	1. Power cycle the device and retry charging process.
DC bus output undervoltage fault	1. Power cycle the device and retry charging process.
DC bus output overcurrent fault	1. Power cycle the device and retry charging process.
Charging module AC input fault	1. Power cycle the device and re-insert the charging connector.
Charging module AC input overvoltage fault	1. Power cycle the device and re-insert the charging connector.
Charging Module AC Input Undervoltage Fault	1. Power cycle the device and re-insert the charging connector.
Charging module AC input phase loss fault	1. Power cycle the device and re-insert the charging connector.
Charging Module DC Output Short Circuit Fault	1. Power cycle the device and retry charging process.
Charging module DC output overcurrent fault	1. Power cycle the device and retry charging process.

Fault	Recommended Action
Charging module DC output overvoltage fault	1. Power cycle the device and retry charging process.
Charging module DC output undervoltage fault	1. Power cycle the device and retry charging process.
Charging module over-temperature fault	1. Wait about half an hour for colling down, then run the charging process for approximately 30 minutes to see if the fault is solved.
Charger cabinet door failure	1. Simulate the door-closed state to see if the problem is solved. 2. Power cycle the device, and close all doors.
DC output contactor adhesion fault	1. While not charging, check output contactor of the charging connector. If the contactor remains closed and there is no voltage at the auxiliary contact, replace the contactor.
Insulation monitoring alarm	1. Power cycle the device and retry charging process.
Drain circuit fault	/
Charging gun over temperature alarm	1. Check whether the resistance of the charging connector temperature sensor matches the PT1000 resistance table. If yes, verify that the wiring is secure. 2. Power cycle the device and run charging process for approximately 30 minutes to see if the fault is solved.
AC input contactor refusal/ misoperation fault	1. Plug in the charging connector into vehicle inlet, then check if the AC contactor makes an operating sound. If not, check if the control circuit functions normally. 2. Power cycle the device and re-insert the charging connector.
AC input contactor adhesion fault	1. Disconnect input voltage and check the continuity of the AC contactor. If the contactor is closed, replace the AC contactor. 2. Power cycle the device.

Fault	Recommended Action
Bridge contactor refusal/ mis-actuation failure	1. While not charging, short the control terminals of the DC contactor. If the contactor does not operate, replace the DC contactor. 2. Power cycle the device and retry charging process.
Bridge contactor adhesion fault	1. While not charging, check the contactor continuity. If the contactor is closed and there is no voltage at the auxiliary contact, replace the DC contactor.
DC charging main control module determines BMS fault	1. Power cycle the device and retry charging process.
BMS sending fault information failure	1. Power cycle the device and retry charging process.
Charging module discharge fault	1. Power cycle the device and retry charging process.
PE wire breakage fault	1. Vrify that the PE input and output wiring is secure.
Control guide voltage abnormal alarm	1. Power cycle the device and retry charging process.
Charging module startup failure alarm	1. Power cycle the device and retry charging process.
Charging module shutdown failure alarm	1. Power cycle the device and retry charging process.
Charging module address abnormal alarm	1. Power cycle the device and retry charging process.
Charging station failed to execute power allocation strategy fault	1. Power cycle the device and retry charging process.
No idle module available fault	1. Power cycle the device and retry charging process.
Pre-charge stage voltage regulation failure	1. Power cycle the device and retry charging process.
Communication timeout between power control modules	1. Power cycle the device and retry charging process.
Charging module 1 fault	1. Power cycle the device and re-insert the charging connector.
Charging module 2 fault	1. Power cycle the device and re-insert the charging connector.

Fault	Recommended Action
Communication failure between charging module 1 and CCU	1. Verify that the CAN communication wiring is secure. 2. Power cycle the device and re-insert the charging connector.
Communication failure between charging module 2 and CCU	1. Verify that the CAN communication wiring is secure. 2. Power cycle the device and re-insert the charging connector.
All charging modules faulty	1. Verify that the CAN communication wiring is secure. 2. Power cycle the device and re-insert the charging connector.
Communication failure between all charging modules and CCU	1. Verify that the CAN communication wiring is secure. 2. Power cycle the device and re-insert the charging connector.
Charging gun A not available	1. Power cycle the device and retry charging process.
Charging gun B not available	1. Power cycle the device and retry charging process.

NOTE

Contact LIVOLTEK for after-sales service if the fault persists or no recommended action is provided.

8.2 Routine Maintenance

8.2.1 Safety Precautions

In order to perform safe and successful maintenance of the device, knowledge of safety precautions must be observed, the necessary tools and test equipment must be used, and qualified maintenance personnel must be involved. Always pay attention to the following safety procedures:

- Dangerous voltages may remain present inside the charger even when it is powered off. Before performing any internal maintenance, wait for at least 10 minutes to allow residual energy to dissipate. Verify with a volmeter that the charger contains no live electricity.
- Do not wear conductive objects such as rings and watches.
- High voltage present inside charger. Without authorization, non-professionals are not allowed to open the charger.

8.2.2 Equipment Maintenance

Table 8-2 Management and Maintenance Test Items

No.	Maintenance Test Items	Period
1	Wear and tear inspection of charging connector connecting cable, check insulation and charging connector for damage.	Season
2	Check input and output terminals to ensure good contact, check terminals for discoloration.	Season
3	Check the inside of the cabinet for standing water or water damage.	Monthly
4	Surface cleaning of cabinets.	Monthly
5	Cleaning the touch screen, check that the display shows normal.	Monthly
6	Check the working condition of the fan to prevent debris from blocking the air outlet.	Monthly
7	Cleaning dust from air inlets and outlets of equipment.	6 months
8	Replacing air filters.	Annual
9	Clean the charging connector for proper use.	Monthly

9 Decommissioning

9.1 Packing

Pack all accessories according to their designated placement directions.

- The sides of the Packing case are printed with warning signs such as fear of moisture, careful and gentle placement, and upward.
- The side of the Packing case is printed with the model number of the equipment and other information.
- The front side of the Packing case is printed with the company logo and the name of the equipment.

9.2 Disposing of the Charger

Properly dispose of the Charger and accessories in accordance with local regulations on the disposal of electronic waste.

10 Technical Specifications

- Charging

Model	MC15	MC20	MC30	MC40	MC60
Charging Mode	Mode 4 EV Chagring				
Rate Output Power [kW]	15	20	30	40	60
Max.Output Current [kW]	15	20	30	40	60
Output Voltage [Vdc]	150 ~ 1000				
Max.Output Current [A]	50	65	100	133	150 or 200 customized
Number of Outputs	1 connector			1 or 2 connector	
Connector Type	CCS2			CCS2/CCS2 + CCS2/CCS2 + CCS1	
Peak Efficiency	≥96%				

- AC Input

Model	MC15	MC20	MC30	MC40	MC60
Earthing System	TN-C | TN-S| TN-C-S| TT				
Input Voltage [Vac]	400 ± 10%(3P + N + PE)				
Input Frenquency [Hz]	50 / 60				
AC Rated Input Current [A]	24	32	48	64	96
AC Max.Input Current [A]	27	36	54	71	108
Power Factor	≥0.99 (@full load)				
THDi	<5%(@full load)				
Min.Connection Cross-section [mm²]	16			35	

- User Interface

Model	MC15	MC20	MC30	MC40	MC60
Status Indication	RGB LED, 10.1' touch screen				
Connectivity	4G, Ethernet 10/100Base-Tx				
Communication Protocols	OCPP 1.61				
User Authentication	APP, RFID card, Credit card (Payter Apollo)				

- Features

Model	MC15	MC20	MC30	MC40	MC60
SPD	SPD II				
RCD	Type A RCD				
OVC.Category	OVC III				
Multiple Protection	Overvoltage Protection, Undervoltage Protection, Overcurrent Protection, Current Leakage Protection, OverTemperature Protection, Grounding Protection, Surge Protection, Short Circuit Protection, Fault Self-check, Insulation Detection and Other Multiple Protection Functions.				
KWH-METER	MID compatible METER				
Environmentat Noise [dB]	< 70 @1m, 25 °C				

- Environmental

Model	MC15	MC20	MC30	MC40	MC60
Operating Temperature [°C]	-30 ~+55 (Derating @50)				
Storage Tmeperature [°C]	-40 ~+70				
Operating Altitude [m]	≤2000				
Humidity	5%- 95% relative				

- Mechanical

Model	MCI5	MC20	MC30	MC40	MC60
IP& IKRating/IP& IK	IP55; IK10				
Casing	SGCC				
Charing Cable Length [m]	5, Standard (including connector) 7, Customized				
Type of installation	Wall mounting or Pole mounting				
Dimensions (W × H× D) [mm]	650 × 800 × 260 (pole: 370 × 722 × 330) (excluding cable management system)				

11 Disclaimer

There are clear requirements and conditions for the transportation, storage, installation and use of this device. LIVOLTEK is not responsible for providing free related service, technical support or compensation in the following cases, including but not limited to:

- The device is beyond its warranty period and no extended warranty has been purchased.
- Failure to provide valid documentation, such as the serial number, warranty card, or purchase invoice.
- Damage resulting from human factors.
- Damage caused by force majeure events (e.g., earthquakes, floods, storms, fires).
- The installation and use of the device violates the relevant local policies and regulations.
- Installation or usage that violates local policies and regulations.
- Installation or usage that does not comply with requirements specified in the user manual.
- Unauthorized modifications to the device's hardware or software.
- Obtaining communication protocols through unauthorized or illegal channels.

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